

DATABASES ASSIGNMENT) Date: _____

Q1) Entity	Attributes	Primary Key
Film	FilmID, Title, Budget, ReleaseDate, DirectorName	FilmID
Actor	ActorID, Name, Phone	ActorID
ProductionCompany	CompanyID, Name	CompanyID

B) Film — Actors
Many to Many
(A film can have many Actors, an Actor can appear in many films)

Film — ProductionCompany
Many - to - Many
A film can be funded by many companies

Film — Director
One to One
~~One~~ One film has one director

c) Casting
Attributes : FilmID, ActorID, Character Name
PK = (FilmID, ActorID)

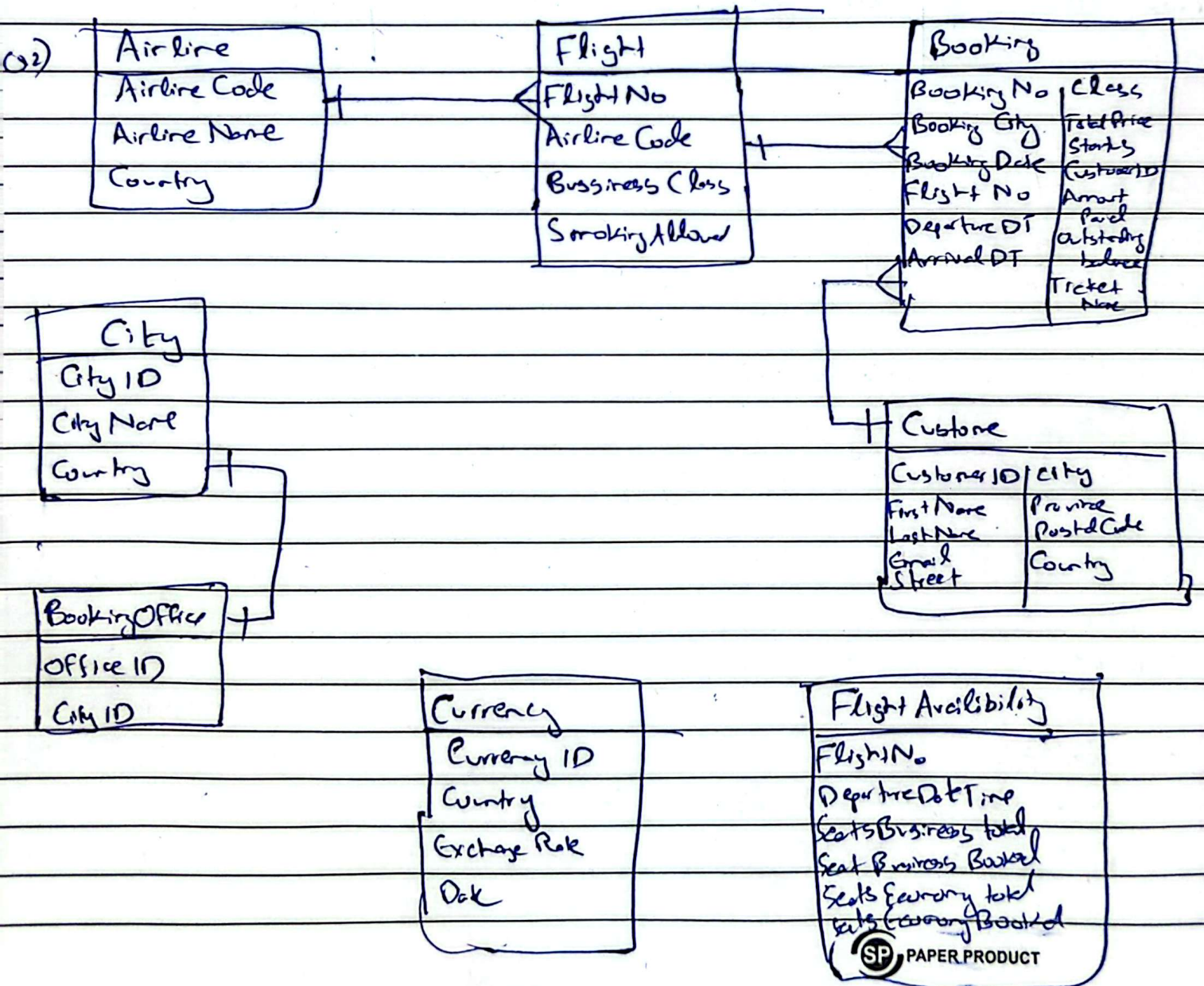
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Funding

Attributes: Film ID, Company ID, Fund Amount

PK = (Film ID, Company ID)

p) Director Name can't be made into a separate entity because it is a non-repeating, descriptive attribute tied directly to a single film.



c)

Date: _____

Q3) Insertion: you can't insert a school unless there is at least one teacher / (you would be required to make dummy data) for example if you want to enter new teacher, you must enter their name, school and contract number

deletion: if you want to delete a teacher contract you'd lose entire data of school as well even though conceptually it would exist

updating: For updating a record for ~~example~~ ~~changing~~ we would have to update school id and school name appearing in multiple rows as leaving even one row would lead to data inconsistency.

Step 1:

b) 1) Functional dependencies

2) $NIN \rightarrow + Name$

(A teacher has unique NIN)

3) $School ID \rightarrow School name, School City$
(School city and school name depends on school ID)

4) $(NIN, contract No, School ID) \rightarrow hours$

Step 2: Convert to 2NF

→ Removing partial dependencies

→ Separating teacher and School information



Date: _____

Teacher :		School		
<u>NIN</u>	<u>tname</u>	<u>SchoolID</u>	<u>SchoolName</u>	<u>SchoolCity</u>
2103				
2103	Brown A	S301	Park School	Manchester
2198	Khan R	S114	Green School	Birmingham
2205	Lewis H			

Converting to 3NF:

Removing transitive dependencies
Attributes fully dependent on key

→ Teacher (NIN, tName)

→ School (SchoolID, SchoolName, SchoolCity)

→ Teacher Contract (NIN, Contract No, SchoolID)

 (NIN) (Contract No) (SchoolID)

Q4) Part A :

(FilmID, ActorID) — Composite Primary Key

b) The table violates 2NF as non key attributes like film title depends on a part (FilmID) and not on the entire composite key. Thus violates 2NF.

→

Date: _____

c) Film (FilmID, FilmTitle, Director Name)

• Dependency Removal: FilmID $\rightarrow$ Director Name, FilmTitle
 $\rightarrow$ Actor (ActorID, Actor Name, AgentID)

• Dependency Removal: ActorID $\rightarrow$ Actor Name, AgentID
 $\rightarrow$ Casting (FilmID, ActorID)

• No partial Dependency now

d) Actor table still contains AgentID and AgentPhone that leads to transitive dependence as AgentID identifies AgentPhone

• Agent $\rightarrow$ AgentID, AgentPhone

• Film $\rightarrow$ FilmID, FilmTitle, DirectorName

• Actor $\rightarrow$ ActorID, AgentID (FK), ActorName

• Casting $\rightarrow$ FilmID, ActorID $\rightarrow$ FK

Both ~~partial~~ partial and transitive dependencies removed

e) Update anomaly as AgentPhone repeats for every film an actor is in hence updating phone may require changing multiple rows that causes inconsistent and contradicting data if even one record is missed.

$\rightarrow$

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Q5. ①

Functional dependencies

Student ID $\rightarrow$ First Name, Last Name, Address

Course ID $\rightarrow$ Course Name, Credits, Instructor Name

(Student ID, Course ID) $\rightarrow$ Enrollment Date

Assumptions: -

- 1) A student is identified by their unique student ID
- 2) A course is identified by unique course ID
- 3) Each course is taught by one instructor only
- 4) A student can ~~enroll~~ enroll in multiple courses.

② table in in 1NF

Student ID | First Name | Last Name | Address | Course ID | Course Name | Credits | Instructor Name | Enrollment Date

remove partial dependencies (2NF)

$\rightarrow$ Student (Student ID, First Name, Last Name, Address)

$\rightarrow$ Course (Course ID, Course Name, Credits, Instructor Name)

$\rightarrow$ Enrollment (Course ID, Student ID, Enrollment Date)

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Transitive dependencies to be removed (3NF)

→ Student (Student ID, First Name, Last Name, Address)

→ Course (Course ID, Course Name, Credits, Instructor Name)

→ Instructor (Instructor Name, Instructor ID)

→ Enrollment (Student ID, Course ID, Enrollment ID)

Composite PK

(3)

Entity	PK	FK	Alternate Key
1) Student	Student ID		
2) Instructor	Instructor ID		
3) Course	Course ID	Instructor Name	Course Name
4) Enrollment	Student ID, Course ID	Student ID, Course ID	